Ziegler spectrum (Ryo Kanda)

§1 Intro
§2 Locally coherent Grothendieck categories
§3 Spectrum of a loc. coh. Groth cats
§4 Ziegler spectrum

§1 Intro.

A : fin. dim alg / k : field.

$$\mathrm{Zg}\, A = \{\text{indec pure-injective right } A\text{-modules}\}/\cong$$
$$\cup$$
$$\{\text{indec fin. dim } A\text{-modules}\}/\cong$$

Thm [Angeleri Hügel – Laking – Sentieri, arXiv:2403.00475]

$$\left\{\begin{array}{c}\text{torsion classes}\\ \text{in } \operatorname{mod} A\end{array}\right\} \overset{1-1}{\longleftrightarrow} \left\{\begin{array}{c}\text{cosilting pairs}\\ (\mathcal{Z}, \mathcal{I})\end{array}\right\}$$

$\mathcal{Z}$: $\operatorname{rg} A$ $\qquad$ $\mathcal{I}$: $\{\text{indec inj}\}/\cong$

$$\cup \qquad\qquad\qquad \cup$$

$$\left\{\begin{array}{c}\text{func. fin.}\\ \text{torsion classes}\end{array}\right\} \overset{1-1}{\underset{[AIR]}{\longleftrightarrow}} \left\{\begin{array}{c}\text{supp } \tau^-\text{-tilting}\\ \text{pairs}\end{array}\right\}$$

$$(\mathcal{C}, \mathcal{I})$$

$\mathcal{C}$: τ^--rigid $\qquad$ $\mathcal{I}$: inj.

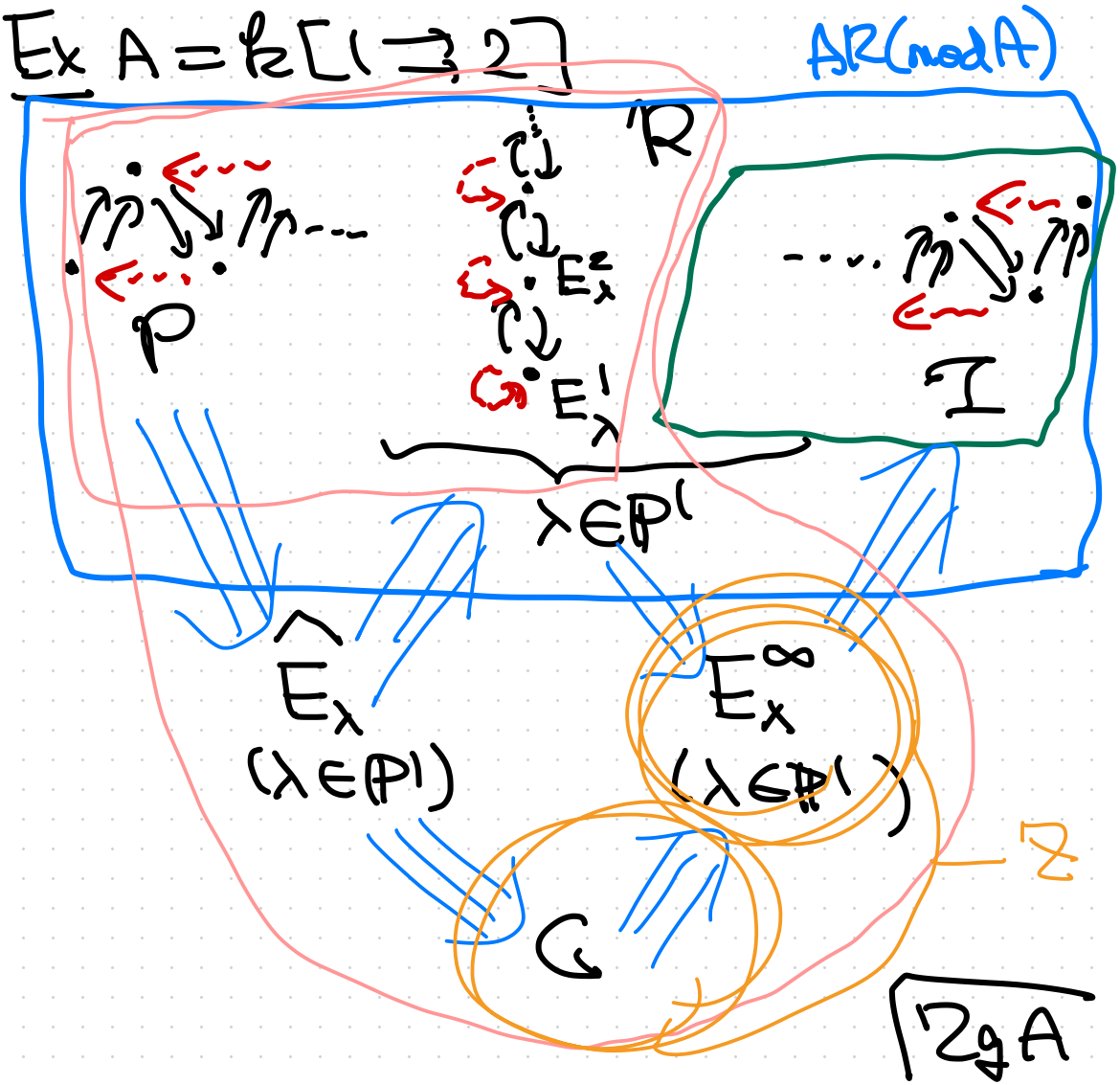
Thm [Crawley-Boevey 1994]

$$\left\{\begin{array}{c}\text{torsion pairs}\\ \text{in } \operatorname{mod} A\end{array}\right\} \overset{1-1}{\longleftrightarrow} \left\{\begin{array}{c}\text{torsion pairs}\\ (\mathcal{T}, \mathcal{F}) \text{ in } \operatorname{Mod} A\\ \text{s.t. } \mathcal{F} \text{ closed under}\\ \text{direct lim } \varinjlim\end{array}\right\}$$

$$(\mathcal{A}, f) \longmapsto \left(\varinjlim \mathcal{A}, \varinjlim f \right)$$

$$\varinjlim \mathcal{A} \;\; \shortparallel \;\; \{\text{direct limits of objs in } \mathcal{A}\}$$

Ex $A = k[1 \rightrightarrows 2]$



$$0 \to E_\lambda^n \to E_\lambda^{n-1} \oplus E_\lambda^{n+1} \to E_\lambda^n \to 0$$

$$\begin{tikzcd} & E_\lambda^{n-1} \\ E_\lambda^n \arrow[ur] \arrow[dr] & \\ & E_\lambda^{n+1} \end{tikzcd}$$

cf. $\begin{cases} \text{[Ringel]} \\ \text{[Prest]} \end{cases}$

$$\varinjlim \left[E_\lambda^1 \hookrightarrow E_\lambda^2 \hookrightarrow E_\lambda^3 \hookrightarrow \cdots \right] =: E_\lambda^\infty$$

Prüfer module.

$$\varprojlim \left[E_\lambda^1 \twoheadleftarrow E_\lambda^2 \twoheadleftarrow E_\lambda^3 \twoheadleftarrow \cdots \right] =: \hat{E}_\lambda$$

adic module.

$$G := \left(k(T) \underset{T}{\overset{1}{\rightrightarrows}} k(T) \right) : \text{generic module.}$$

These are all the indec pure-inj.

§2 Locally coherent Groth cats

<u>Motivation</u>

loc. coh. Groth cat

$$\underset{\text{ring}}{R} \rightsquigarrow (\mathrm{mod}(R^{op}), Ab) \xrightarrow[Sp]{} \mathcal{D}_g R$$

$$\underset{\text{compactly generated triangulated cat}}{\mathcal{T}} \rightsquigarrow ((\mathcal{T}^c)^{op}, Ab) \xrightarrow[Sp]{} \mathcal{D}_g \mathcal{T}$$

$$\left(\begin{array}{l} \mathcal{C}, \mathcal{D} : \mathbb{Z}\text{-linear cats} \\ (\mathcal{C}, \mathcal{D}) := \{ \mathcal{C} \to \mathcal{D} : \text{additive functors} \} \end{array} \right)$$

Def $\mathcal{G}$: Grothendieck cat

$:\Longleftrightarrow$

- $\mathcal{G}$: abelian cat
- $\mathcal{G}$ has coproducts
  ($=:$ direct sum $\oplus$)
  ($\rightsquigarrow$ $\mathcal{G}$ has all colimits)
- direct lim $\varinjlim$ is exact.
  (For a direct system of short ex sequences, its termwise direct lim is again exact.)
- $\exists U \in \mathcal{G}$: generator,
  (that is, $\forall M \in \mathcal{G}$, $\exists U^{\oplus} \underset{\text{epi}}{\twoheadrightarrow} M$.)

$\left(\Rightarrow \mathcal{G}\right.$ has all limits,

inj cogenerator,

$\left.\text{inj envelopes.}\right)$

<u>Ex</u> R: ring $\Rightarrow$ Mod R: Groth cat.

I: small $\mathbb{Z}$-linear cat

$\Rightarrow (I, Ab)$: Groth cat.

$\mathcal{G}$: Groth cat.

<u>Def</u> ① $M \in \mathcal{G}$: finitely generated

$:\Leftrightarrow \left[\begin{array}{l} M = \sum_{i \in I} L_i, \quad L_i \subset M \text{ subobject} \\ \Rightarrow M = L_{i_1} + \cdots + L_{i_n} \\ \quad \exists i_1, \ldots, i_n \in I \end{array}\right]$

$\Longleftrightarrow \mathrm{Hom}(M,-) : \mathcal{G} \to \mathrm{Ab}$

preserves $\underbrace{\text{direct unions}}$.

direct limit of direct system whose transition mors are monomorphisms.

② $M \in \mathcal{G}$: finitely presented

$:\Longleftrightarrow \begin{cases} \cdot\ M : \text{fin. gen.} \\ \cdot \left[\begin{array}{l} 0 \to K \to \underset{\text{fg}}{L} \to M \to 0 \\ \Longrightarrow K : \text{fg.} \end{array} \right] \end{cases}$

$\Longleftrightarrow \mathrm{Hom}(M,-) : \mathcal{G} \to \mathrm{Ab}$

preserves $\varinjlim$.

Def $M \in \mathcal{G}$: coherent

$$:\Longleftrightarrow \begin{cases} \cdot\ M : \text{fin. pres.} \\ \cdot \left[\underset{\text{fg}}{L} \subset M \Rightarrow L : \text{fp} \right]. \end{cases}$$

Rmk coh $\Rightarrow$ fp $\Rightarrow$ fg $\Rightarrow$ compact

(Hom(M, −) preserves direct sums.)

Def $\mathcal{G}$: locally coherent

(finitely presented (fg), noeth, ...)

$:\Longleftrightarrow$ Every obj in $\mathcal{G}$ is a direct limit of coherent obj's (fp, noeth, ...)

<u>Ex</u> R: ring.

$\mathrm{Mod}\,R$ is always locally fp.

$\mathrm{Mod}\,R$ is locally coherent (noeth)

$\Longleftrightarrow$ R: right coherent (noeth)

<u>Thm</u> [Roos 1969, Auslander 1971?]

$\mathcal{G}$: locally fp. TFAE:

① $\mathcal{G}$: locally coherent.

② $\mathrm{fp}\,\mathcal{G} = \mathrm{coh}\,\mathcal{G}$.

③ $\mathrm{fp}\,\mathcal{G}$ is closed under kernels. (hence wide)

④ $\mathrm{fp}\,\mathcal{G}$ is an abelian cat.

Def $M \in \mathcal{C}$: noetherian

$$:\Longleftrightarrow \left[\begin{array}{l} L_0 \subset L_1 \subset \cdots \subset M \\ \Rightarrow \exists i,\ L_i = L_{i+1} = \cdots \end{array} \right].$$

Rmk $\cancel{\left(\text{noeth} \overset{?}{\Rightarrow} \text{coh}\right) \Rightarrow \cdots}$ noeth $\not\Rightarrow$ fp

$\Downarrow$ fg

loc noeth $\Rightarrow$ loc coherent.

$\Uparrow$ $\qquad\qquad$ $\Downarrow$

loc of fin. length $\qquad$ loc fp

$\Downarrow$

loc fg

Thm [Roos 1969]

Every obj is noeth.

$$\left\{\begin{array}{c}\text{loc coherent (noeth)}\\ \text{Groth cats}\end{array}\right\}/\cong \;\overset{1-1}{\longleftrightarrow}\; \left\{\begin{array}{c}\text{small (noeth)}\\ \text{abelian cats}\end{array}\right\}/\cong$$

$$\begin{array}{ccc} \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ \mathcal{G} & \longmapsto & \mathrm{fp}\,\mathcal{G} \\ \mathrm{Lex}(\mathcal{A}^{op}, \mathrm{Ab}) & \longleftarrow\!\shortmid & \mathcal{A} \\ \rotatebox{90}{$\cong$} & & \\ \{\mathcal{A}^{op} \to \mathrm{Ab} : \text{left exact}\} & & \end{array}$$

$\mathrm{Hom}(-, M)$ $\quad$ $M \in \mathcal{A}$

Prop [Auslander 1971?]

I: small additive cat.

(I, Ab) is always locally finitely presented Groth cat.

$F \in (I, \mathrm{Ab})$: fp

$\iff \mathrm{Hom}(M, -) \to \mathrm{Hom}(L, -) \to F \to 0$ $\quad$ ($\exists\, L \to M$ in I.)

TFAE:

① $(\mathcal{I}, Ab)$ is locally coherent

② $\mathcal{I}$ has pseudo-cokernels,

that is,

$\forall f : L \to M$ in $\mathcal{I}$, $\exists\, M \to N$,

$\mathrm{Hom}(N, -) \to \mathrm{Hom}(M, -) \to \mathrm{Hom}(L, -)$

is exact.

<u>Ex</u> R: ring. $\mathrm{mod}\,R \subset \mathrm{Mod}\,R$

$\quad\quad\quad

$\mathcal{T}$: tri cat

cone is pseudo-cokernel.

$$L \xrightarrow{f} M \longrightarrow \underbrace{N}_{\text{pseudo-cok of } f.} \longrightarrow L[1]$$

<u>Def</u> A Serre subcat of an abelian cat is a full subcat closed under subobjs, quotient objs, and extensions.

<u>Def</u> $\mathcal{G}$: Grothendieck cat.

① $\mathcal{X} \underset{\text{full}}{\subset} \mathcal{G}$: localizing

$:\Longleftrightarrow \mathcal{X}$: Serre, closed under $\oplus$

$\Longleftrightarrow \mathcal{X} \hookrightarrow \mathcal{Y}$ has right adj.

$\wedge$

$\mathcal{X}$: Serre,

$\Longleftrightarrow \mathcal{X}$: Serre, $\mathcal{Y} \twoheadrightarrow \frac{\mathcal{Y}}{\mathcal{X}}$

has a right adj.

② $\mathcal{X} \subset \mathcal{Y}$: localizing.

$\mathcal{X}$ is of finite type

$:\Longleftrightarrow$ The right adj $\mathcal{Y} \to \mathcal{X}$

preserves $\underrightarrow{\lim}$.

$\Longleftrightarrow$ The right adj $\frac{\mathcal{Y}}{\mathcal{X}} \to \mathcal{Y}$

preserves $\underrightarrow{\lim}$.

<u>Thm</u> Let $\mathcal{G}$: loc coh Groth cat.

$(\leftrightarrow \mathrm{fp}\,\mathcal{G})$

① $\left\{ \begin{array}{l} \mathcal{X} \subset \mathcal{G} : \text{loc} \\ \quad \text{of finite type} \end{array} \right\} \overset{1-1}{\leftrightarrow} \left\{ \begin{array}{c} \mathcal{S} \subset \mathrm{fp}\,\mathcal{G} \\ \text{Serre} \end{array} \right\}$

$$\begin{array}{ccc} \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ \mathcal{X} & \longmapsto & \mathcal{X} \cap \mathrm{fp}\,\mathcal{G} \\ \varinjlim \mathcal{S} & \longleftarrow\!\shortmid & \mathcal{S} \end{array}$$

② Let $\mathcal{X} \subset \mathcal{G}$: loc of fin. type and $\mathcal{S} := \mathcal{X} \cap \mathrm{fp}\,\mathcal{G}$.

Then $\mathcal{X}$, $\dfrac{\mathcal{G}}{\mathcal{X}}$ are loc coh Groth cats.

$$\mathrm{fp}\,\mathcal{X} = \mathcal{S}, \quad \mathrm{fp}\,\frac{\mathcal{G}}{\mathcal{X}} = \frac{\mathrm{fp}\,\mathcal{G}}{\mathcal{S}}.$$

§3 Spectrum of a locally coherent Grothendieck category

$\mathcal{G}$: locally coherent Groth cat

<u>Def</u> $\mathrm{Sp}\,\mathcal{G} := \{$indecomposable injective objs in $\mathcal{G}\}/\cong$

: the <u>spectrum</u> of $\mathcal{G}$.

<u>Def</u> For $C \in \mathrm{fp}\,\mathcal{G}$.

$$\mathcal{O}(C) := \{ I \in \mathrm{Sp}\,\mathcal{G} \mid \mathrm{Hom}(C, I) \neq 0 \}$$

$\{\mathcal{O}(C) \mid C \in \mathrm{fp}\,\mathcal{G}\}$ satisfies the axiom of a basis of open subsets of $\mathrm{Sp}\,\mathcal{G}$.

$\rightsquigarrow$ Define a topology on $\mathrm{Sp}\,\mathcal{G}$.

<u>Rmk</u> · An inj obj in $\mathcal{C}$ may not have indec decomp.

· $\bigoplus$ inj may not be inj.

<u>Prop</u> ① $\mathcal{O}(C) = \emptyset \Leftrightarrow C = 0$.

② $\mathcal{O}(C)$ is compact (and open).

All open cpt subsets of $\mathrm{Sp}\,\mathcal{C}$ are of this form.

<u>Proof</u> ① Suppose $0 \neq C \in \mathrm{fp}\,\mathcal{C}$.

$\exists\, \underbrace{L}_{\substack{\text{maximal} \\ \text{(Zorn's lemma)} \\ (=: C:\mathrm{fg})}} \subsetneq C \twoheadrightarrow \underbrace{\frac{C}{L} = S}_{\text{simple.}} \hookrightarrow \underbrace{E(S)}_{\substack{\text{indec} \\ \text{inj.}}}$

$E(S) \in \mathcal{O}(<)$.

//

<u>Ex</u> R : comm noeth ring. Then

$$\begin{array}{ccc} \operatorname{Spec} R & \overset{1-1}{\longleftrightarrow} & \operatorname{Sp}(\operatorname{Mod} R) \\ \cup & & \cup \\ \mathfrak{p} & \longmapsto & E_R(R/\mathfrak{p}) \\ \Phi & \longleftrightarrow & \cup \\ \vdots & & \vdots \end{array}$$

specialization-closed $\Longleftrightarrow$ open

closed $V(I)$ $\Longleftrightarrow$ open cpt

<u>Thm</u> [Herzog 1997, Krause 1997]

$\exists$ order-preserving bijs:

$$\begin{tikzcd}
\left\{ \begin{array}{c} \mathcal{X} \subset \mathcal{G} : \text{loc} \\ \text{of fin. type} \end{array} \right\} \arrow[rr, leftrightarrow, "1-1"] \arrow[d, leftrightarrow, "1-1"'] & & \left\{ \begin{array}{c} \mathcal{S} \subset \mathrm{fp}\,\mathcal{G} \\ \text{Serre} \end{array} \right\} \arrow[d, leftrightarrow, "1-1"] \\
\{ U \subset \mathrm{Sp}\,\mathcal{G} : \text{open} \} \arrow[rr, leftrightarrow] & & \left\{ \begin{array}{c} C \in \mathrm{fp}\,\mathcal{G} \mid \\ \Theta(C) \subset U \end{array} \right\}
\end{tikzcd}$$

$$\mathcal{S} \longmapsto \Theta(\mathcal{S}) := \bigcup_{C \in \mathcal{S}} \Theta(C)$$

$$U \longmapsto \left\{ C \in \mathrm{fp}\,\mathcal{G} \mid \Theta(C) \subset U \right\}$$

$$\mathrm{Sp}\,\mathcal{X} = \Theta(\mathcal{S})$$

$$\mathrm{Sp}\,\frac{\mathcal{G}}{\mathcal{X}} = \mathrm{Sp}\,\mathcal{G} \setminus \Theta(\mathcal{S}).$$

<u>Ex</u> k : field.

$\mathrm{cmod}\, k := \{$vector sp of dim $\leq \aleph_0\}$ $\subset$ Mod k

$\subset$

mod k

cmod k "countable dim"

$$\mathrm{cmod}\, k \xrightarrow{Q} \mathcal{A} := \frac{\mathrm{cmod}\, k}{\mathrm{mod}\, k} \longleftrightarrow \mathcal{G} : \text{loc coh Noeth cat.}$$

$\mathcal{A}$ has only two isoclasses $0, M = Q(k^{\oplus \aleph_0})$

two Serre subcats $0, \mathcal{A}$.

$E(M)$ in $\mathcal{G}$ is <u>superdecomposable</u>

$\Updownarrow$

nonzero & has no indec summand.

But $\exists S \in \mathcal{G}$: simple.

$E(S)$: indec inj.

$\therefore$ $E(S)$ has no fp subobj.

$\mathcal{A} = \mathrm{fp}\,\mathcal{G}$

$\mathrm{fp}\,\mathcal{G}$. in $\mathcal{G}$

$\not\exists\, L \subset M \twoheadrightarrow \frac{M}{L} =: S$.

max subobj

$\in \mathrm{fp}\,\mathcal{G}$.

If

$\mathrm{fp} \subset E(S)$, then

$\neq 0$

$M \subset E(S)$

$\rightsquigarrow E(M) = E(S)$. $\lightning$.

superdecomp

indec

§4 Ziegler spectrum of a module category. (ring)

Let R: ring. $\operatorname{mod} R = \{\text{f.p. right mods}\}$

$\operatorname{mod} R^{op} := \{\text{fin pres left } R\text{-mods}\}$.

$\subset \operatorname{Mod} R$

closed under cok.

$\rightsquigarrow (\operatorname{mod} R^{op}, Ab)$: loc coh Grothcat.

$\rightsquigarrow Sp(\operatorname{mod} R^{op}, Ab)$.

<u>Thm</u> [Gruson–Jensen 1973, ...]

See Prest.

① $\varepsilon : \operatorname{Mod} R \to (\operatorname{mod} R^{op}, Ab)$: fully faithful.

$\perp$

$$M \longmapsto M \otimes_R -$$

$$F(R) \longleftarrow\!\!\shortmid\ F$$

② $\operatorname{Im}\varepsilon = \{ F \in (\operatorname{mod} R^{op}, Ab)$ right ex $\}$

$\cup$

$\operatorname{Inj}(\operatorname{mod} R^{op}, Ab)$

$\Updownarrow$

absolutely pure

$\Updownarrow$

$\operatorname{Ext}^1(C, F) = 0,$

$\forall C \in \operatorname{fp}(\operatorname{mod} R^{op}, Ab).$

③ $\operatorname{Pure\,Inj} R \xrightarrow{\sim} \operatorname{Inj}(\operatorname{mod} R^{op}, Ab)$

all pure-inj mods (defined below) — all inj functors.

Def-Prop

① $0 \to L \xrightarrow{\text{pure-mono}} M \to N \xrightarrow{\text{pure-epi}} 0$ ex in $\mathrm{Mod}R$

is <u>pure-exact</u>

$:\Longleftrightarrow 0 \to L \otimes_R - \to M \otimes_R - \to N \otimes_R - \to 0$

is exact in $(\mathrm{mod}R^{op}, \mathrm{Ab})$.

$\Longleftrightarrow$ ——— // ———

is exact on $\mathrm{Mod}R^{op}$.

$\Longleftrightarrow {}^{\forall} C \in \mathrm{mod}R$,

$0 \to \mathrm{Hom}(C,L) \to \mathrm{Hom}(C,M) \to \mathrm{Hom}(C,N) \to 0$

: ex in Ab.

$\Longleftrightarrow 0 \to L \to M \to N \to 0$ is a direct limit of split short exact seqs.

② $N \in \mathrm{Mod}\,R$: <u>pure-inj</u>

$:\Longleftrightarrow$ Every pure exact

$$0 \to N \to M \to L \to 0$$

splits.

Π pure-inj is pure-inj.

<u>Def</u> $\mathrm{Zg}\,R := \{$indec pure-inj right R-mods$\}/\cong$.

$$\left(\underset{|-|}{\xrightarrow{\Sigma}} \mathrm{Sp}(\mathrm{mod}\,R^{op}, \mathrm{Ab}) \right).$$

The topology on Sp induces a topology on $\mathrm{Zg}\,R$.

<u>Prop</u> $\mathrm{Zg}\,R$ is compact.

<u>Proof</u> $R \otimes_R - \in \mathrm{Sp}(\mathrm{mod}\,R^{op}, \mathrm{Ab})$.

$$\mathcal{O}(R \otimes_R -) = \mathrm{Sp}(\mathrm{mod}\,R^{op}, \mathrm{Ab}) \;//$$

<u>Ex</u> R: comm ring, A: R-alg.

(not nec module-finite!)

$E \in \mathrm{Mod}\,R$: inj. $M \in \mathrm{Mod}\,A$.

$$\Longrightarrow \begin{cases} \cdot\ M^* := \mathrm{Hom}_R(M, E) \in \mathrm{Mod}\,A^{op} \\ \qquad \text{is always pure-inj.} \\ \cdot\ M \hookrightarrow M^{**} : \text{pure-mono} \end{cases}$$

<u>Ex</u> K: field, A: K-alg.

All fin. dim A-module is pure-inj.

$(\because)$ $M^{**} \cong M$ is pure-inj. //

<u>Thm</u> [Auslander 1986, Gruson–Jensen 1981]

$d : \mathrm{fp}(\mathrm{mod}\, R, \mathrm{Ab}) \to \mathrm{fp}(\mathrm{mod}\, R^{op}, \mathrm{Ab})$: contravariant equivalence.

$F \in \mathrm{fp}(\mathrm{mod}\, R, \mathrm{Ab})$

$$(dF)(\underset{\mathrm{mod}\, R^{op}}{L}) := \mathrm{Hom}(F, - \otimes_R L)$$

$$(d^{-1}G)(C) := \mathrm{Hom}(G, C \otimes_R -).$$

- $d(- \otimes_R L) = \mathrm{Hom}_R(L, -)$.
- $d(\mathrm{Hom}_R(C, -)) = C \otimes_R -$.

$(L \in \mathrm{mod}\, R^{op},\ C \in \mathrm{mod}\, R.)$

Prop $\forall F \in (\mathrm{mod}R, \mathrm{Ab})$,

$\exists \vec{F} \in (\mathrm{Mod}R, \mathrm{Ab})$

preserving $\varinjlim$,

$\vec{F}|_{\mathrm{mod}R} \cong F$.

$\vec{F}$ is unique up to $\cong$.

Proof $\vec{F}(\varinjlim_i \underset{\substack{\uparrow \\ \mathrm{fp}}}{M_i}) := \varinjlim_i F(M_i)$.

$$\underbrace{\bigoplus_i \mathrm{Hom}(\underset{\substack{\uparrow \\ \mathrm{fp}}}{C_i}, -) \to \bigoplus_j \mathrm{Hom}(\underset{\substack{\uparrow \\ \mathrm{fp}}}{D_j}, -)}_{\text{defined on } \mathrm{Mod}R} \to F \to 0$$

//

Observation

$$\begin{tikzcd} \mathrm{Zg}R \arrow[r, "\varepsilon", "(-)"'] & \mathrm{Sp}(\mathrm{mod}R^{op}, \mathrm{Ab}) \\ \exists U(F) \arrow[u, phantom, "\cup"] \arrow[r, "\sim"'] & \mathcal{O}(F) \arrow[u, phantom, "\cup"] \end{tikzcd}$$

$N \in \mathrm{Zg}R.$ $\quad N = \varinjlim_i C_i \ (C_i \in \mathrm{mod}R).$

$$\begin{aligned} N \in U(F) &\iff \varepsilon N \in \mathcal{O}(F) \\ &\iff \mathrm{Hom}(F, N \otimes_R -) \neq 0 \\ &\iff \varinjlim_i \mathrm{Hom}(F, C_i \otimes_R -) \neq 0 \\ &\iff \varinjlim_i (dF)(C_i) \neq 0 \\ &\iff (\overrightarrow{dF})(\underbrace{\varinjlim_i C_i}_{= N}) \neq 0 \\ &\iff (\overrightarrow{dF})(N) \neq 0. \end{aligned}$$

$\therefore\ U(F) = \{ N \in \mathrm{Zg}R \mid (\overrightarrow{dF})(N) \neq 0 \}.$

$U'(G) := \{ N \in \mathrm{Zg}R \mid \overrightarrow{G}(N) \neq 0 \}.$

$G \in \mathrm{fp}(\mathrm{mod}R, \mathrm{Ab})$

$\{ U'(G) \mid G \in \mathrm{fp}(\mathrm{mod}R, \mathrm{Ab}) \}$

is an open basis of $\mathrm{Zg}R$.

$\mathrm{Zg}R \setminus U'(G) = \mathrm{Zg}R \cap \underbrace{\mathrm{Ker}\,\overrightarrow{G}}_{\subseteq \mathrm{Mod}R}$

Arbitrary closed subset is an intersection of this form.

Def $\mathcal{Z} \underset{\text{full}}{\subset} \operatorname{Mod} R$: <u>definable</u>

$:\Longleftrightarrow \exists \{F_i\}_{i \in I} \subset \operatorname{fp}(\operatorname{mod} R, \mathrm{Ab})$,

$$\mathcal{Z} = \bigcap_{i \in I} \operatorname{Ker} \overrightarrow{F_i}.$$

(cf. pp-pair).

$\Longleftrightarrow \mathcal{Z}$ is closed under

$\prod$, $\varinjlim$, pure-submods.

($\Rightarrow \bigoplus$, $\langle \oplus$, pure-quot mods,

pure-inj envelopes).

Rmk

$M \underset{\operatorname{fp}}{\otimes}_R -$ preserves $\prod$, $\varinjlim$.

<u>Thm</u> (See Prest Cor 12.3.6)

$$\{Z \subset \mathrm{Mod}\,R : \text{definable}\} \overset{1-1}{\longleftrightarrow} \{X \subset \mathrm{Zg}\,R\}$$

$$\begin{array}{ccc} \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ Z & \longmapsto & \mathrm{Zg}\,R \cap Z \\ \langle X \rangle & \longleftarrow\!\shortmid & X \\ \text{definable} & & \text{closed} \end{array}$$

§5 Ziegler spectrum of a compactly generated triangulated cat.

$\mathcal{T}$: tri cat with coproducts.

$\mathcal{T}^c := \{\text{cpt objs in } \mathcal{T}\}$.

<u>Def</u> $\mathcal{T}$: compactly generated

$\Leftrightarrow \exists \mathcal{C} \subset \mathcal{T}^c$: set,

$\forall X \in \mathcal{T}$,

$[\mathrm{Hom}(\mathcal{C}, X) = 0 \Rightarrow X = 0]$.

$\Leftrightarrow \exists \mathcal{C} \subset \mathcal{T}^c$: set,

$\langle \mathcal{C} \rangle_{\text{localizing}} = \mathcal{T}$.

full tri subcats closed under coproducts.
(thick)

$\mathcal{T}$: cpt gen tri cat.

Ex $\mathcal{T} = D(\mathrm{Mod}\,R)$. $\mathcal{T}^c = K^b(\mathrm{proj}\,R)$.
cpt gen. tri cat.

phantom.

Def ① $X \to Y \to Z \to X[1]$: tri in $\mathcal{T}$

pure-mono pure-epi

is <u>pure-exact</u>

$:\Longleftrightarrow \forall C \in \mathcal{T}^c$,

$$0 \to \mathrm{Hom}(C, X) \to \mathrm{Hom}(C, Y) \to \mathrm{Hom}(C, Z) \to 0$$

: exact.

$\Longleftrightarrow \forall C \in \mathcal{T}^c$,

$\mathrm{Hom}(C, Z) \to \mathrm{Hom}(C, X[1])$ is zero.

i.e. $Z \to X[1]$ is a <u>phantom morphism</u>.

$\Longleftrightarrow$ This is a homology colimit of a filter diagram of split triangles.

② $N \in T$: pure-inj

$:\Longleftrightarrow$ Every pure exact tri $N \to M \to L \to N[1]$ splits.

$\Longleftrightarrow$ Only phantom mor to N is 0.

<u>Def</u> $\mathrm{Zg}\, T := \{$indec pure-inj objs in $T\}/\cong$.

Rmk $\mathcal{T} \xrightarrow{y} ((\mathcal{T}^c)^{op}, Ab)$: not full.

$$\begin{array}{ccc} \mathcal{T} & \xrightarrow{y} & ((\mathcal{T}^c)^{op}, Ab) \\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ X & \longmapsto & \mathrm{Hom}(-, X) \end{array}$$

locally coh Groth cat.

$$\mathrm{fp}((\mathcal{T}^c)^{op}, Ab)$$
$$= \{ F \in ((\mathcal{T}^c)^{op}, Ab) \mid \mathrm{Hom}(-, C_1) \to \mathrm{Hom}(-, C_0) \to F \to 0 \}$$
$$\exists\, C_1 \to C_0 \text{ in } \mathcal{T}^c.$$

Thm [Krause 2000]

① $N \in \mathcal{T}$: pure-inj

$\iff \forall M \in \mathcal{T}$,

$$\mathrm{Hom}(M, N) \xrightarrow[y]{\sim} \mathrm{Hom}(yM, yN).$$

② $\mathrm{PureInj}\, T \xrightarrow[\mathcal{F}]{\sim} \mathrm{Inj}((T^c)^{op}, Ab)$

$\therefore \mathrm{Zg}\, T \overset{1-1}{\longleftrightarrow} \mathrm{Sp}((T^c)^{op}, Ab)$.

Top on Sp induces top on Zg.

<u>Thm</u> [Krause 2002]

$\exists$ bij b.w.

① closed subsets $\subset \mathrm{Zg}\, T$.

② definable subcats $\subset T$.

$\uparrow$ $\bigcap_{i \in I} \mathrm{Ker}\, \overrightarrow{F_i}$,

$F_i \in \mathrm{fp}((T^c)^{op}, Ab)$.

R : ring. $T = D(\mathrm{Mod}\,R)$.

<u>Thm</u> [Garkusha–Prest 2005].

① $X \in D(\mathrm{Mod}\,R)$: pure-inj

$\Rightarrow {}^\forall n \in \mathbb{Z},\ H^n(X) \in \mathrm{Mod}\,R$ is pure-inj.

② $N \in \mathrm{Mod}\,R$: pure-inj

$\Rightarrow {}^\forall n \in \mathbb{Z},\ N[n] \in D(\mathrm{Mod}\,R)$ is pure-inj.

③ Suppose R is right hereditary.

$\{\text{pure-inj in } D(\mathrm{Mod}\,R)\}$

$= \left\{ \bigoplus_{n \in \mathbb{Z}} N_n[n] \;\middle|\; N_n \in \mathrm{Mod}\,R : \text{pure-inj} \right\}$

$$\therefore \operatorname{Zg} D(\operatorname{Mod} R) = \bigcup_{n \in \mathbb{Z}} (\operatorname{Zg} R)[n].$$